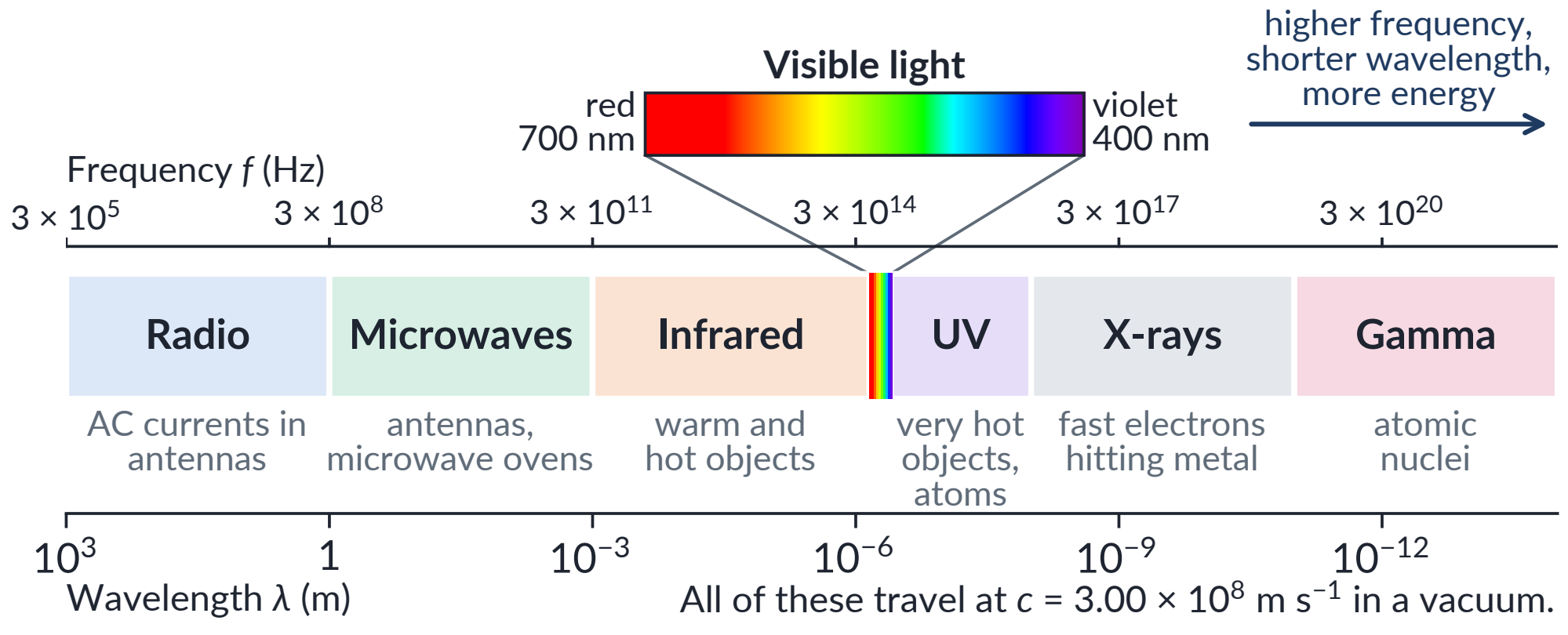
Visible light
red
700 nm
violet
400 nm
higher frequency,
shorter wavelength,
more energy
Frequency f (Hz)
3 × 10^5
3 × 10^8
3 × 10^11
3 × 10^14
3 × 10^17
3 × 10^20
Radio
Microwaves
Infrared
UV
X-rays
Gamma
AC currents in antennas
antennas, microwave ovens
warm and hot objects
very hot objects, atoms
fast electrons hitting metal
atomic nuclei
10^3
1
10^−3
10^−6
10^−9
10^−12
Wavelength λ (m)
All of these travel at c = 3.00 × 10^8 m s^−1 in a vacuum.